

How to use these notes: read each part once, then jump to the QUICK REVISION SHEET at the very end. The numericals are solved step by step, each followed by my own similar example.

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PART 0: SLIDE 1 and 2 (Title and roadmap)

Slide 1 is just the title page plus a copyright note. The authors (Kurose and Ross) let you use the slides freely, but you must credit them as the source. All material is copyright 1996 to 2025.

Slide 2: chapter goal. The goal is to get a "feel" and the big picture, plus learn the terminology. The deeper details come later in the course.

The roadmap of the chapter (and the order of these notes):

1. What is the Internet? What is a protocol?
2. Network edge: hosts, access networks, physical media
3. Network core: packet switching, circuit switching, Internet structure
4. Performance: loss, delay, throughput
5. Protocol layers and service models
6. Security
7. History

Note: the slide roadmap lists Performance, then Protocol layers, then Security, then History. The lecture actually goes Performance, then Security, then Layers, then History. Not a big deal, just do not get confused.

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PART 1: WHAT IS THE INTERNET? (Slides 3 to 6)

There are TWO ways the slides look at the Internet. Remember both:

- (a) the "nuts and bolts" view: what the Internet is made of
- (b) the "services" view: what the Internet does for us

1.1 NUTS AND BOLTS VIEW (Slide 3)

The Internet has these components:

8. Hosts (also called end systems)
 - Billions of connected computing devices.
 - They run network applications at the Internet's "edge".
 - Examples: phones, laptops, desktops, servers, cars, smart fridges, light bulbs.
1. Packet switches
 - They forward packets (chunks of data).

- Examples: routers and switches.
- 1. Communication links
 - Made of fiber, copper, radio, satellite.
 - Each link has a transmission rate, which is called bandwidth.
- 1. Networks
 - A collection of devices, routers and links that is managed by an organization.
 - Types shown on the slide: mobile network, home network, enterprise network, content provider network, datacenter network, local or regional ISP, national or global ISP.

Simple way to remember: Host = the thing you use. Link = the road. Router/switch = the road junction that decides where each packet goes next.

1.2 FUN INTERNET-CONNECTED DEVICES (Slide 4)

The slide shows that not only computers are on the Internet. Examples listed:

- Tweet-a-watt (monitors energy use)
- Internet refrigerator
- IP picture frame
- Amazon Echo
- Sensorized bed mattress
- Pacemaker and monitor
- AR devices
- Fitbit
- Bikes, cars, scooters
- Web-enabled toaster plus weather forecaster
- Slingbox (remote control of cable TV)
- Internet phones
- Security camera
- Gaming devices
- "Others?" (the slide asks you to think of more)

Point of the slide: the Internet of Things is huge, anything can become an end system.

1.3 NETWORK OF NETWORKS AND PROTOCOLS (Slide 5)

- The Internet is a "network of networks": many interconnected ISPs (Internet Service Providers).
- Protocols are everywhere. They control the sending and receiving of messages.
- Examples of protocols: HTTP (Web), streaming video, Zoom, TCP, IP, WiFi, 4G/5G, Ethernet.
- Internet standards: written as RFCs (Request for Comments) and produced by the IETF (Internet Engineering Task Force).

Memory trick: IETF is the group, RFC is the document they write.

1.4 SERVICES VIEW (Slide 6)

- The Internet is an infrastructure that provides services to applications. Examples: Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, connected appliances.
- It provides a programming interface to distributed applications. These are "hooks" that let sending and receiving apps "connect" to and use the Internet transport service.
- It provides service options, analogous to the postal service.

Easy example: when you post a letter you do not care how the post office moves it. You just drop it in the box and choose a service (normal or courier). Similarly an app developer just "hands data" to the Internet through an interface and chooses a service type. The Internet does the rest.

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PART 2: WHAT IS A PROTOCOL? (Slides 7 and 8)

Human protocols: asking "what's the time?", saying "I have a question", introductions.

Network protocols: the same idea, but between computers (devices) instead of humans. ALL communication activity on the Internet is governed by protocols.

DEFINITION (very important, write this in exams):

Protocols define the FORMAT and ORDER of messages sent and received among network entities, and the ACTIONS taken on message transmission and receipt.

So a protocol has three things:

1. Format of messages
2. Order of messages
3. Actions taken when a message is sent, received or some other event occurs

Slide 8 example (compare a human and a network protocol):

Human: you say "Hi", the other says "Hi", you ask "Got the time?", they reply "2:00".

Computer: the client sends a "TCP connection request", the server sends a "TCP connection response", then the client sends "GET http://gaia.cs.umass.edu/kurose_ross", then the server sends back the <file>.

Time flows downward in these diagrams. The notes on the slide say you will see this kind of diagram a lot, showing distributed entities exchanging messages governed by protocols.

Slide question: "Other human protocols?" Examples you can write: greeting when picking up a phone call ("Hello?" then "Hi, is this...?"), a classroom where you raise your hand before asking a question, a bank teller queue, ordering food at a restaurant.

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PART 3: NETWORK STRUCTURE OVERVIEW (Slides 9 to 12)

Slide 9 is the roadmap again. Slides 10, 11 and 12 build the picture step by step. The Internet has three parts:

4. NETWORK EDGE (Slide 10)
 - Hosts: clients and servers.
 - Servers are often in data centers.
1. ACCESS NETWORKS AND PHYSICAL MEDIA (Slide 11)
 - Wired and wireless communication links that connect end systems to the first router.
1. NETWORK CORE (Slide 12)
 - Interconnected routers.
 - A network of networks.

Easy picture: Edge = your house and devices. Access = the road from your house to the main road. Core = the highways between cities.

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PART 4: ACCESS NETWORKS AND PHYSICAL MEDIA (Slides 13 to 21)

Slide 13 question: How do we connect end systems to the edge router? Three kinds:

- Residential access nets
- Institutional access networks (school, company)
- Mobile access networks (WiFi, 4G/5G)

4.1 CABLE-BASED ACCESS (Slides 14 and 15)

- Uses the existing cable TV network.
- Uses frequency division multiplexing (FDM): different channels are transmitted in different frequency bands. Some channels carry video (TV), some carry data, some carry control.
- At the home: cable modem and splitter.
- At the cable company: cable headend with a CMTS (Cable Modem Termination System), which connects to the ISP.
- Data and TV travel at different frequencies over a SHARED cable distribution network.
- HFC = Hybrid Fiber Coax. A network of cable and fiber attaches homes to the ISP router.

- Homes SHARE the access network to the cable headend (so speed can drop when neighbors all use it).
- Asymmetric: downstream 40 Mbps to 1.2 Gbps, upstream 30 to 100 Mbps. Downstream is faster than upstream.

4.2 DSL: DIGITAL SUBSCRIBER LINE (Slide 16)

- Uses the EXISTING TELEPHONE LINE to a central office.
- Has a DSL modem and splitter at home, and a DSLAM (DSL Access Multiplexer) at the central office.
- Voice and data travel at different frequencies over a DEDICATED line to the central office.
- Data goes to the Internet through the ISP. Voice goes to the telephone network.
- Downstream 24 to 52 Mbps (dedicated), upstream 3.5 to 16 Mbps (dedicated).

Cable vs DSL difference (very common question):

- Cable: SHARED medium among homes. DSL: DEDICATED line for each home.
- Cable uses TV cable. DSL uses telephone line.
- Cable headend/CMTS vs. DSL central office/DSLAM.

4.3 HOME NETWORKS (Slide 17)

- Has wireless and wired devices.
- The cable or DSL modem, the router, firewall and NAT are often combined in a single box.
- WiFi wireless access point: 54 Mbps or 450 Mbps.
- Wired Ethernet: 1 Gbps.
- The connection goes to/from the headend or central office.

4.4 WIRELESS ACCESS NETWORKS (Slide 18)

A shared wireless access network connects an end system to a router via a base station, also called an "access point".

Two types:

1. Wide-area cellular access networks
 - Provided by a mobile cellular network operator.
 - Range: tens of km.
 - Speed: tens of Mbps.
 - Examples: 4G/5G.
1. Wireless Local Area Networks (WLANs)
 - Typically within or around a building (about 100 ft).
 - 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rates.

4.5 PHYSICAL MEDIA (Slides 19, 20, 21)

First the vocabulary:

- A bit propagates between transmitter/receiver pairs.
- The physical link is what lies between transmitter and receiver.
- Guided media: signals travel in solid media (copper, fiber, coax).
- Unguided media: signals propagate freely, for example radio.

A) Twisted Pair (TP)

- Two insulated copper wires.
- Category 5: 100 Mbps and 1 Gbps Ethernet.
- Category 6: 10 Gbps Ethernet.

B) Fiber optic cable

- Glass fiber that carries light pulses, each pulse is a bit.
- High speed point-to-point transmission: tens to hundreds of Gbps.
- Low error rate: repeaters are spaced far apart.
- Immune to electromagnetic noise.

C) Coaxial cable

- Two concentric copper conductors.
- Bidirectional.
- Broadband: multiple frequency channels on the cable, hundreds of Mbps per channel.

D) Wireless radio

- Signal is carried in various "bands" in the electromagnetic spectrum.
- No physical "wire".
- Broadcast and "half-duplex" (sender to receiver).
- Propagation environment effects: reflection, obstruction by objects, interference/noise.
- Radio link types:
 - o Wireless LAN (WiFi): 10 to 100s of Mbps, range of tens of meters.
 - o Wide-area (4G/5G cellular): hundreds of Mbps over about 10 km.
 - o Bluetooth: cable replacement, short distances, limited rates.
 - o Terrestrial microwave: point-to-point, 45 Mbps channels.
 - o Satellite: up to less than 100 Mbps (Starlink) downlink. About 270 msec end-to-end delay for geostationary satellites (because they are very far up).

Memory trick for physical media speed ranking (slowest to fastest, roughly): Bluetooth/WiFi < Coax (100s Mbps) < Twisted pair Cat6 (10 Gbps) < Fiber (10s to 100s Gbps).

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PART 5: THE NETWORK CORE (Slides 22 to 34)

5.1 PACKET SWITCHING (Slide 23)

- The core is a mesh of interconnected routers.
- Packet switching: hosts break application-layer messages into packets.
- The network forwards packets from one router to the next, across links on the path from source to destination.

5.2 TWO KEY CORE FUNCTIONS (Slides 24, 25, 26)

1. ROUTING
 - GLOBAL action.
 - Determines the source-to-destination paths taken by packets.
 - Uses routing algorithms.
1. FORWARDING (also called "switching")
 - LOCAL action.
 - Moves an arriving packet from the router's input link to the appropriate router output link.
 - Uses a local forwarding table.

How forwarding works on the slide: a packet arrives with a destination address in its header, for example 0111. The router looks this up in its local forwarding table:

Header value : Output link

0100 : 3

0101 : 2

0111 : 2

1001 : 1

The packet with 0111 is sent out on link 2.

Analogy (slides 25 and 26 use map pictures): ROUTING is like planning your whole trip on a map (choosing the path from your city to the destination). FORWARDING is like going through one interchange and choosing the correct exit right now. Routing is planning, forwarding is acting.

5.3 HOST SENDING FUNCTION (Slide 27)

The host:

- takes the application message
- breaks it into smaller chunks called packets of length L bits
- transmits the packet into the access network at transmission rate R

R is called the link transmission rate, or link capacity, or link bandwidth.

PACKET TRANSMISSION DELAY = L / R

(time needed to transmit an L-bit packet into the link). L is in bits, R is in bits per second, so the answer is in seconds.

5.4 STORE AND FORWARD (Slide 28)

- Takes L/R seconds to transmit (push out) an L-bit packet into a link of R bps.
- Store and forward: the ENTIRE packet must arrive at the router before it can be transmitted on the next link.

NUMERICAL ON THE SLIDE (one hop):

Given: $L = 10$ Kbits, $R = 100$ Mbps.

Step 1: convert to the same units. $L = 10$ Kbits = 10,000 bits. $R = 100$ Mbps = 100,000,000 bits/sec.

Step 2: delay = $L/R = 10,000 / 100,000,000 = 0.0001$ sec.

Step 3: 0.0001 sec = 0.1 msec.

Answer: one-hop transmission delay = 0.1 msec. (This matches the slide.)

MY SIMILAR EXAMPLE 1:

A packet of 8000 bits goes onto a link of 2 Mbps. Find the transmission delay.

$L = 8000$ bits, $R = 2,000,000$ bps.

Delay = $8000 / 2,000,000 = 0.004$ sec = 4 msec.

Extension (not on the slide but the classic follow-up): with store and forward, if a packet crosses N links each of rate R (and we ignore propagation, queueing, processing), the end-to-end delay is $N \times L/R$.

MY EXAMPLE 2: 3 links, $L = 10$ Kbits, $R = 100$ Mbps each. Total = 3×0.1 msec = 0.3 msec.

Why? Because each router must receive the whole packet before sending it, so the packet is fully "pushed" onto each of the 3 links, one after another.

5.5 QUEUEING AND LOSS (Slides 29 and 30)

Slide setup: hosts A and B send at $R = 100$ Mb/s into a router. The router's output link to C, D, E is $R = 1.5$ Mb/s. There is a queue of packets waiting for transmission over the output link.

- Queueing occurs when work arrives faster than it can be serviced.
- If the arrival rate (bps) to a link exceeds the transmission rate (bps) of the link for some period of time:
 1. packets will queue, waiting to be transmitted on the output link
 2. packets can be dropped (lost) if the memory (buffer) in the router fills up

Analogy: a funnel. You pour water (100 Mb/s) faster than the narrow neck (1.5 Mb/s) can let it through. Water waits in the funnel (queue). If the funnel overflows, water is lost (packet loss).

5.6 CIRCUIT SWITCHING (Slide 31)

This is the ALTERNATIVE to packet switching.

- End-to-end resources are allocated to and RESERVED for a "call" between source and destination.
- In the diagram each link has four circuits. The call gets the 2nd circuit in the top link and the 1st circuit in the right link.
- Dedicated resources: no sharing.
- Circuit-like (guaranteed) performance.
- A circuit segment is IDLE if not used by the call (no sharing, so it is wasted).
- Commonly used in traditional telephone networks.

5.7 FDM and TDM (Slide 32)

Two ways to split a link into circuits:

1. Frequency Division Multiplexing (FDM)
 - The optical/electromagnetic frequencies are divided into narrow frequency bands.
 - Each call gets its own band, and can transmit at the max rate of that NARROW band (all the time).
 - Slide example: 4 users, each with their own frequency band.
1. Time Division Multiplexing (TDM)
 - Time is divided into slots.
 - Each call gets periodic slot(s) and can transmit at the maximum rate of the (WIDER) frequency band, but ONLY during its time slot(s).

Analogy: FDM is like 4 people each using their own lane on a highway all the time. TDM is like 4 people taking turns using the whole road, each for a few seconds.

5.8 PACKET SWITCHING VS CIRCUIT SWITCHING (Slides 33 and 34)

NUMERICAL ON THE SLIDE:

Given: a 1 Gb/s link. Each user sends at 100 Mb/s when "active". Each user is active only 10% of the time. There are N users.

Question: how many users can use this network under circuit switching and under packet switching?

Circuit switching:

- Each user needs a reserved 100 Mb/s, even when idle.
- Number of users = link capacity / rate per user = $1000 \text{ Mb/s} / 100 \text{ Mb/s} = 10$ users.
Answer: 10 users.

Packet switching:

- Users only use the link when active, and on average only 10% of the time.
- Say 35 users. The link can handle up to 10 active users at once ($10 \times 100 \text{ Mb/s} = 1 \text{ Gb/s}$).
- The chance that MORE than 10 of the 35 are active at the same time is tiny. The slide says it is less than 0.0004. (I checked with the binomial formula and got about 0.00042, so it is roughly 0.0004, which is essentially the same thing.)
- So packet switching supports 35 users (more than 3 times as many) with almost no chance of overload.

Where does the probability come from? Each user is active with probability $p = 0.1$, independently. The number of active users among N follows a binomial distribution. $P(\text{more than 10 active}) = \sum_{k=11}^{35} C(35,k) \times (0.1)^k \times (0.9)^{(35-k)}$.

MY SIMILAR EXAMPLE 3:

A 1 Mbps link. Each user needs 100 kbps when active and is active 20% of the time.

- Circuit switching: $1 \text{ Mbps} / 100 \text{ kbps} = 10$ users.
- Packet switching with 20 users: average active users = $20 \times 0.2 = 4$. Overload only happens if MORE than 10 are active at once. The binomial calculation ($N = 20, p = 0.2$) gives a probability of about 0.0006, which is tiny.
- So packet switching supports 20 users, double the circuit-switching count, and the network is overloaded only about 0.06% of the time.
- Lesson: the burstier the users (the lower the activity percentage), the bigger the gain from packet switching. Try 30 users: the probability becomes about 0.026 (2.6%), so it is getting riskier. The more users you add, the higher the overload chance.

Slide 34: Is packet switching a "slam dunk winner"?

Pros of packet switching:

- Great for "bursty" data (sometimes you have data to send, sometimes you do not).
- Resource sharing.
- Simpler, no call setup.

Cons of packet switching:

- Excessive congestion is possible: packet delay and loss due to buffer overflow.
- Protocols are needed for reliable data transfer and congestion control.

Slide questions:

- Q: How to provide circuit-like behavior with packet switching? Answer on slide: "It's complicated." Later chapters cover techniques that try to make packet switching as circuit-like as possible.

- Q: Human analogies of reserved resources (circuit switching) vs on-demand allocation (packet switching)?
Good answers: circuit switching is like making a restaurant reservation (table is held for you even if you are late). Packet switching is like walking in and waiting for any free table. Another: a private car/taxi reserved for you vs a shared bus.

Quick comparison table in words:

Circuit switching: reserved, guaranteed performance, wasteful when idle, needs call setup, used in old telephone networks.

Packet switching: shared, no guarantee, efficient for bursty data, no setup, can have delay and loss, used in the Internet.

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PART 6: INTERNET STRUCTURE: A NETWORK OF NETWORKS (Slides 35 to 43)

Slide 35 basics:

- Hosts connect to the Internet via access ISPs.
- Access ISPs must be interconnected so that any two hosts anywhere can send packets to each other.
- The resulting network of networks is very complex.
- Its evolution is driven by economics and national policies.
- The slides build the current structure STEP BY STEP. Remember the steps:

STEP 1 (Slides 36 and 37): Given millions of access ISPs, how do we connect them?

Option: connect every access ISP to every other directly. This does NOT scale. It needs $O(N^2)$ connections. (With N ISPs, about $N \times N/2$ links. Example: 1000 ISPs would need nearly 500,000 links.)

STEP 2 (Slide 38): Connect each access ISP to ONE global transit ISP.

- Customer and provider ISPs have an economic agreement (the customer pays the provider).

STEP 3 (Slide 39): If one global ISP is a viable business, there will be competitors (ISP A, ISP B, ISP C).

STEP 4 (Slide 40): Competitors will want to be connected to each other. This happens through:

- Internet Exchange Points (IXPs)
- Peering links (direct connections between two ISPs, where they usually exchange traffic without paying each other)

STEP 5 (Slide 41): Regional networks may arise to connect access nets to ISPs (regional ISP).

STEP 6 (Slide 42): Content provider networks (for example Google, Microsoft, Akamai) may run their OWN network to bring services and content close to end users.

FINAL STRUCTURE (Slide 43):

At the "center" is a small number of well-connected large networks:

- "Tier-1" commercial ISPs (examples: Level 3, Sprint, AT&T, NTT), with national and international coverage.
- Content provider networks (examples: Google, Facebook): a private network connecting their data centers to the Internet, often BYPASSING tier-1 and regional ISPs.
Below them: regional ISPs, then access ISPs. IXPs connect them.

Memory trick for the hierarchy: Tier-1 (top) > Regional ISP > Access ISP > your home. Content providers are the "rich shortcut" that skip the middle.

PART 7: PERFORMANCE: DELAY, LOSS, THROUGHPUT (Slides 44 to 54)

7.1 HOW DELAY AND LOSS OCCUR (Slide 45)

- Packets queue in router buffers, waiting their turn for transmission.
- The queue length grows when the arrival rate to the link (temporarily) exceeds the output link capacity.
- Packet loss occurs when the memory to hold queued packets fills up.
- Diagram parts: the packet being transmitted (transmission delay), packets in buffers (queueing delay), free buffers (arriving packets are dropped if there are no free buffers).

7.2 FOUR SOURCES OF PACKET DELAY (Slides 46 and 47)

Total nodal delay:

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

1. d_{proc} (nodal processing delay)
 - Check bit errors, determine the output link.
 - Typically less than microseconds.
1. d_{queue} (queueing delay)
 - Time waiting at the output link for transmission.
 - Depends on the congestion level of the router.
1. d_{trans} (transmission delay)
 - L = packet length (bits), R = link transmission rate (bps).

- $d_{trans} = L / R$
1. d_{prop} (propagation delay)
 - d = length of the physical link, s = propagation speed (about 2×10^8 m/sec).
 - $d_{prop} = d / s$

BIG POINT: d_{trans} and d_{prop} are VERY DIFFERENT.

- d_{trans} is about how fast you can push the bits OUT onto the link (depends on packet size and link rate).
- d_{prop} is about how long it takes a bit to TRAVEL along the link (depends on distance and speed).

Memory trick: Transmission = Time to put bits on the wire. Propagation = Time for bits to Pass through the wire (travel).

MY EXAMPLE 4 (computing all four delays):

A host sends a 1500-byte packet over a 10 Mbps link that is 2000 km long. Processing delay is 0.05 msec and queueing delay is 2 msec. Find the nodal delay.

- $L = 1500 \text{ bytes} \times 8 = 12,000 \text{ bits}$. $R = 10,000,000 \text{ bps}$.
- $d_{trans} = 12,000 / 10,000,000 = 0.0012 \text{ sec} = 1.2 \text{ msec}$.
- $d_{prop} = 2,000,000 \text{ m} / (2 \times 10^8 \text{ m/s}) = 0.01 \text{ sec} = 10 \text{ msec}$.
- $d_{proc} = 0.05 \text{ msec}$, $d_{queue} = 2 \text{ msec}$.
- $d_{nodal} = 0.05 + 2 + 1.2 + 10 = 13.25 \text{ msec}$.

Notice that propagation is the biggest piece here because the link is long. On a short link with a slow rate, transmission would dominate instead.

7.3 CARAVAN ANALOGY (Slides 48 and 49)

Mapping:

- car = bit
- caravan = packet (ten-car caravan = 10-bit packet)
- toll booth = link (or router, in slide 49)
- toll service = link transmission
- driving on the highway = propagation

NUMERICAL 1 (Slide 48):

Given: a ten-car caravan. The toll booth takes 12 sec to service one car (bit transmission time). Cars propagate at 100 km/hr. The toll booths are 100 km apart.

Question: how long until the caravan is lined up before the 2nd toll booth?

Step 1: time to "push" the entire caravan through the first toll booth onto the highway = 12 sec x 10 cars = 120 sec (this is the transmission delay).

Step 2: time for the last car to travel from the first toll booth to the second = $100 \text{ km} / (100 \text{ km/hr}) = 1 \text{ hr} = 60 \text{ min}$ (this is the propagation delay).

Step 3: total = $120 \text{ sec} + 60 \text{ min} = 2 \text{ min} + 60 \text{ min} = 62 \text{ minutes}$.

Answer: 62 minutes (matches the slide).

MY SIMILAR EXAMPLE 5:

An eight-car caravan. Toll booth takes 10 sec per car. Cars travel at 120 km/hr. Booths are 60 km apart. When is the whole caravan lined up at the second booth?

- Push time = $10 \times 8 = 80 \text{ sec}$.
- Travel time = $60 / 120 = 0.5 \text{ hr} = 30 \text{ min} = 1800 \text{ sec}$.
- Total = $80 + 1800 = 1880 \text{ sec} = 31 \text{ min } 20 \text{ sec}$.

NUMERICAL 2 (Slide 49):

Now suppose cars propagate at 1000 km/hr and the toll booth takes ONE MIN to service a car.

Question: Will cars arrive at the 2nd booth before all cars are serviced at the first booth?

Step 1: the first car is finished at the first booth after 1 min.

Step 2: it travels 100 km at 1000 km/hr = $0.1 \text{ hr} = 6 \text{ min}$.

Step 3: it arrives at the second booth at $1 + 6 = 7 \text{ min}$.

Step 4: after 7 min, the first booth has serviced 7 cars. There are 10 cars, so $10 - 7 = 3$ cars are STILL waiting at the first booth.

Answer: YES. After 7 min the first car arrives at the second booth while three cars are still at the first booth. (Matches the slide.)

What this means in networking: the first bit of a packet can reach the next router BEFORE the last bit has even been sent by the first router. This happens when transmission is slow (big d_{trans}) and propagation is fast (small d_{prop}).

MY SIMILAR EXAMPLE 6:

Ten-car caravan. Cars travel at 600 km/hr. Booths are 100 km apart. Toll takes 2 min per car. Do cars reach booth 2 before all are serviced at booth 1?

- First car leaves booth 1 at 2 min.
 - Travel = $100 / 600 \text{ hr} = 1/6 \text{ hr} = 10 \text{ min}$.
 - Arrives at booth 2 at $2 + 10 = 12 \text{ min}$.
 - By 12 min, booth 1 has serviced $12 / 2 = 6$ cars, so $10 - 6 = 4$ cars are still at booth 1.
- Answer: YES, 4 cars still at the first booth.

7.4 QUEUEING DELAY REVISITED (Slide 50)

Variables:

- a = average packet arrival rate (packets/sec)

- L = packet length (bits)
- R = link bandwidth (bits/sec)

Traffic intensity = $L \times a / R$
 (arrival rate of bits) divided by (service rate of bits).

What it means:

- La/R close to 0: average queueing delay is small.
- La/R going toward 1: average queueing delay becomes large.
- La/R greater than 1: more work is arriving than can be serviced, so the average delay is INFINITE (the queue just keeps growing).

The graph: average queueing delay on the vertical axis and traffic intensity on the horizontal axis. It stays flat and low at first, then shoots up steeply as La/R approaches 1. The key golden rule: DESIGN YOUR SYSTEM SO THAT TRAFFIC INTENSITY IS NOT CLOSE TO 1.

There is no fixed numerical on the slide, so here are examples:

MY EXAMPLE 7:

Packets are 12,000 bits long (1500 bytes). Link rate $R = 10$ Mbps. The arrival rate is $a = 500$ packets/sec.

- $La = 12,000 \times 500 = 6,000,000$ bits/sec.
- Traffic intensity = $6,000,000 / 10,000,000 = 0.6$.
- The link is 60% utilized, so the queueing delay is moderate.

Now $a = 900$ packets/sec:

- $La = 10,800,000$ bits/sec.
- Traffic intensity = 1.08, which is greater than 1.
- Bits arrive faster than the link can send them, so the queue grows without limit and delay becomes infinite (in reality the buffer fills up and packets are lost).

7.5 PACKET LOSS (Slide 51)

- The queue (buffer) before a link has FINITE capacity.
- A packet arriving at a full queue is dropped (lost).
- A lost packet may be retransmitted by the previous node, by the source end system, or not at all.

7.6 THROUGHPUT (Slides 52, 53, 54)

Definitions:

- Throughput = the rate (bits per time unit) at which bits are being sent from sender to receiver.
- Instantaneous throughput: the rate at a given point in time.
- Average throughput: the rate over a longer period of time.

Slide 52 pipe analogy: a server with a file of F bits sends bits (like fluid) into a pipe. The first link (server side) has capacity R_s bits/sec and the second link (client side) has capacity R_c bits/sec.

Slide 53:

- If $R_s < R_c$, the average end-to-end throughput = R_s (the smaller one).
- If $R_s > R_c$, the average end-to-end throughput = R_c (the smaller one).
- In both cases, throughput = $\min(R_s, R_c)$.
- The bottleneck link is the link on the end-to-end path that constrains the end-to-end throughput.

Analogy: a water pipe made of two sections, a thick one and a thin one. The water that comes out the end is limited by the thin section.

Slide 54 (network scenario):

- 10 connections (fairly) share a backbone bottleneck link of R bits/sec. Each connection has its own R_s on the server side and R_c on the client side.
- Per-connection end-to-end throughput = $\min(R_c, R_s, R/10)$.
- In practice, R_c or R_s is often the bottleneck (the backbone R is usually huge).

MY EXAMPLE 8 (two links):

Server link $R_s = 2$ Mbps, client link $R_c = 1$ Mbps. Throughput = $\min(2, 1) = 1$ Mbps. The bottleneck is the client link.

How long to send a file of $F = 8$ Mbits? Time = $F / \text{throughput} = 8 / 1 = 8$ sec.

MY EXAMPLE 9 (shared backbone, like slide 54):

$R_s = 1$ Mbps, $R_c = 2$ Mbps, backbone $R = 5$ Mbps shared by 10 connections.

- $R/10 = 0.5$ Mbps.
- Throughput = $\min(2, 1, 0.5) = 0.5$ Mbps.
- The BACKBONE is the bottleneck here.
- Time to send a file of 4 Megabytes (= 32 Mbits) = $32 / 0.5 = 64$ sec.
Remember to convert bytes to bits (x 8).

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PART 8: NETWORK SECURITY (Slides 55 to 61)

Slides 56 and 57 (same content, repeated):

- The Internet was NOT originally designed with much security in mind.
- The original vision was "a group of mutually trusting users attached to a transparent network".
- Internet protocol designers are playing "catch-up".
- Security considerations are needed in ALL layers.

We now need to think about:

1. how bad guys can attack computer networks
2. how we can defend networks against attacks
3. how to design architectures that are immune to attacks

BAD GUY ATTACK 1: PACKET INTERCEPTION (Slide 58)

- Called packet "sniffing".
- Works on broadcast media (shared Ethernet, wireless).
- A promiscuous network interface reads and records ALL packets passing by, including passwords.
- Example in the slide: B sends a packet to A (src:B dest:A payload), but C, who is also on the same medium, reads it.
- Wireshark (used in the end-of-chapter labs) is a free packet sniffer.

BAD GUY ATTACK 2: FAKE IDENTITY (Slide 59)

- IP spoofing: injecting a packet with a FALSE source address.
- Example: C sends a packet to A but writes "src:B" so A thinks it came from B.

BAD GUY ATTACK 3: DENIAL OF SERVICE, DoS (Slide 60)

- Attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming the resource with bogus traffic.
- Three steps:
 1. Select a target.
 2. Break into hosts around the network (this creates a botnet).
 3. Send packets to the target from the compromised hosts.
- When many hosts attack together it is usually called distributed DoS.

LINES OF DEFENSE (Slide 61)

- Authentication: proving you are who you say you are. Cellular networks provide hardware identity via the SIM card. There is no such hardware assist in the traditional Internet.
- Confidentiality: via encryption.
- Integrity checks: digital signatures prevent or detect tampering.
- Access restrictions: password-protected VPNs.
- Firewalls: specialized "middleboxes" in access and core networks. They are off-by-default, and filter incoming packets to restrict senders, receivers, applications. They also help with detecting and reacting to DoS attacks.
- More on security comes later (Chapter 8).

Quick trick to remember the attacks: SNIFF (steal by listening), SPOOF (pretend to be someone), DoS (flood until it breaks).

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PART 9: PROTOCOL LAYERS AND REFERENCE MODELS (Slides 62 to 72)

9.1 WHY LAYERING? (Slides 63 and 64)

- Networks are complex with many pieces: hosts, routers, links of various media, applications, protocols, hardware, software.
- Question: is there any hope of organizing the structure of a network and our discussion of networks? Answer: yes, by layering.
- Explicit structure allows identification and relationship of the system's pieces (a layered reference model for discussion).
- Modularization eases maintenance and updating of the system.
- A change in one layer's service implementation is TRANSPARENT to the rest of the system. The slide's example: a change in the gate procedure at an airport does not affect the rest of the airline system.

9.2 THE INTERNET PROTOCOL STACK (Slide 65): FIVE LAYERS

From top to bottom:

1. APPLICATION: supporting network applications. Protocols: HTTP, IMAP, SMTP, DNS.
2. TRANSPORT: process-to-process data transfer. Protocols: TCP, UDP.
3. NETWORK: routing of datagrams from source to destination. Protocols: IP, routing protocols.
4. LINK: data transfer between NEIGHBORING network elements. Protocols: Ethernet, 802.11 (WiFi), PPP.

5. PHYSICAL: bits "on the wire".

Memory trick (top to bottom): "All People Seem To Need Data Processing" is the classic OSI one. For this 5-layer stack try: "A Teacher Never Lets Play" (Application, Transport, Network, Link, Physical). Make up your own if that is weird.

Key distinction to remember:

- Transport = between PROCESSES (apps) on two hosts.
- Network = between HOSTS across the whole path.
- Link = between NEIGHBORS (one hop).

9.3 ENCAPSULATION (Slides 66 to 71)

Each layer adds its own header to the data coming from the layer above. This is called encapsulation. The names of the data units change at each layer:

- Application layer: MESSAGE (M)
- Transport layer: SEGMENT = [Ht | M] (Ht is the transport header)
- Network layer: DATAGRAM = [Hn | Ht | M] (Hn is the network header)
- Link layer: FRAME = [Hl | Hn | Ht | M] (Hl is the link header)

Step by step (source host):

1. The application exchanges a message M to implement an application service, using the services of the transport layer.
2. The transport layer protocol transfers M (e.g., reliably) from one process to another using the services of the network layer. It ENCAPSULATES M with a transport header Ht to create a transport-layer segment. Ht is used by the transport layer protocol to implement its service.
3. The network layer protocol transfers the segment [Ht | M] from one host to another using link layer services. It encapsulates the segment with a network header Hn to create a network-layer datagram. Hn is used by the network layer protocol to implement its service.
4. The link layer protocol transfers the datagram from host to neighboring host using network-layer services (as the slide words it). It encapsulates the datagram with a link-layer header Hl to create a link-layer frame.
5. The physical layer sends the bits.

At the destination the reverse happens: each layer removes (strips) its own header and passes the rest up.

Slide 69 analogy: Matryoshka (stacking) dolls. Message is the smallest doll, which goes inside the segment doll, which goes inside the datagram doll, which goes inside the frame doll. Put dolls inside bigger dolls, then open them one by one at the other side.

Real life analogy: you write a letter (message), put it in an envelope with a department address (segment), put that in a bigger courier envelope with the city address (datagram), and the delivery van adds a route sticker for the next depot (frame).

Slide 71: end-to-end view

- The SOURCE and DESTINATION hosts implement all five layers.
- A ROUTER implements three layers: network, link, physical. It looks at the network header (Hn) to decide where to forward the packet.
- A (link-layer) SWITCH implements two layers: link and physical. It looks at the link header (Hl).
- Headers are added and removed hop by hop at the link layer, while the network and transport headers (and the message) stay the same from end to end (the router only reads the network layer part, it does not open the transport part).

Memory trick: Router = up to layer 3 (network). Switch = up to layer 2 (link). Hosts = all the way to layer 5 (application).

9.4 ISO/OSI REFERENCE MODEL (Slide 72)

- This is the SEVEN layer model: application, presentation, session, transport, network, link, physical.
- The two layers NOT found in the Internet stack are:
 - Presentation: allows applications to interpret the meaning of data, e.g., encryption, compression, machine-specific conventions.
 - Session: synchronization, checkpointing, recovery of the data exchange.
- The Internet stack is "missing" these layers. These services, if needed, must be implemented in the application. The slide asks "needed?" (leaving you to think about whether they really were needed).

Internet stack (5 layers) vs OSI (7 layers): the extra two are Presentation and Session, which sit between Application and Transport.

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(UNIMPORTANT)—> PART 10: INTERNET HISTORY (Slides 73 to 78)

PERIOD 1: 1961 to 1972, early packet-switching principles (Slide 74)

- 1961: Kleinrock, queueing theory shows the effectiveness of packet switching.
- 1964: Baran, packet switching in military nets.
- 1967: ARPAnet conceived by the Advanced Research Projects Agency.
- 1969: first ARPAnet node operational.
- 1972: ARPAnet public demo. NCP (Network Control Protocol) is the first host-host protocol. First e-mail program. ARPAnet has 15 nodes.

PERIOD 2: 1972 to 1980, internetworking and new/proprietary networks (Slide 75)

- 1970: ALOHAnet satellite network in Hawaii.
- 1974: Cerf and Kahn, an architecture for interconnecting networks.
- 1976: Ethernet at Xerox PARC.
- Late 70s: proprietary architectures DECnet, SNA, XNA.
- 1979: ARPAnet has 200 nodes.
- Cerf and Kahn's internetworking principles (these define today's Internet architecture):
 1. Minimalism, autonomy: no internal changes are required to interconnect networks.
 2. Best-effort service model.
 3. Stateless routing.
 4. Decentralized control.

PERIOD 3: 1980 to 1990, new protocols and a proliferation of networks (Slide 76)

- New national networks: CSnet, BITnet, NSFnet, Minitel.
- 100,000 hosts connected to a confederation of networks.
- 1982: SMTP e-mail protocol defined.
- 1983: deployment of TCP/IP.
- 1983: DNS defined for name-to-IP-address translation.
- 1985: FTP protocol defined.
- 1988: TCP congestion control.

PERIOD 4: 1990 and the 2000s, commercialization, the Web, new applications (Slide 77)

- Early 1990s: ARPAnet decommissioned.
- 1991: NSF lifts restrictions on the commercial use of NSFnet (NSFnet decommissioned in 1995).
- Early 1990s: the Web. Hypertext [Bush 1945, Nelson 1960s]. HTML and HTTP by Berners-Lee.
- 1994: Mosaic, later Netscape.
- Late 1990s: commercialization of the Web.
- Late 1990s to 2000s: more "killer apps" (instant messaging, P2P file sharing), network security comes to the forefront, about 50 million hosts and 100 million+ users, backbone links running at Gbps.

PERIOD 5: 2005 to present, scale, SDN, mobility, cloud (Slide 78)

- Aggressive deployment of broadband home access (10s to 100s of Mbps).
- 2008: software-defined networking (SDN).
- Increasing ubiquity of high-speed wireless access: 4G/5G, WiFi.
- Service providers (Google, Facebook, Microsoft) create their own networks, bypassing the commercial Internet to connect "close" to the end user, providing "instantaneous" access to social media, search, video content.
- Enterprises run their services in the "cloud" (e.g., Amazon Web Services, Microsoft Azure).
- Rise of smartphones: more mobile than fixed devices on the Internet (2017).
- About 15 billion devices attached to the Internet (2023, statista.com).

Memory trick for the dates: 1969 = first ARPAnet node. 1974 = Cerf and Kahn. 1976 = Ethernet. 1983 = TCP/IP and DNS. 1991 = commercial use allowed. 2008 = SDN. 2017 = mobile devices overtake fixed.

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PART 11: CHAPTER SUMMARY (Slide 79)

What was covered: Internet overview, what a protocol is, network edge/access network/core, packet switching vs circuit switching, Internet structure, performance (loss, delay, throughput), layering and service models, security, history.

What you now have: context, overview, vocabulary and a "feel" of networking. More depth comes later.

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QUICK REVISION SHEET (read this the day before the exam)

DEFINITIONS

- Host = end system. Packet switch = router or switch. Link = fiber, copper, radio, satellite.
- Protocol = format + order of messages + actions taken on send/receive.
- RFC = document, IETF = group that makes Internet standards.
- Bandwidth = link transmission rate (bits per second).
- Bottleneck link = the slowest link on the path that limits throughput.

FORMULAS

- Transmission delay: $d_{\text{trans}} = L / R$ (bits divided by bits per second).
- Propagation delay: $d_{\text{prop}} = d / s$ (distance divided by speed, s is about 2×10^8 m/s).

- Nodal delay: $d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$.
- Store-and-forward over N links (ignoring other delays): $N \times L / R$.
- Traffic intensity = $L \times a / R$. Near 0 is small delay, near 1 is big delay, above 1 means infinite delay.
- Throughput = $\min(R_s, R_c)$ for two links. With a shared backbone of 10 connections: $\min(R_c, R_s, R/10)$.
- Circuit switching users = link rate / rate per user.
- Number of links if every ISP connects to every other: about N squared ($O(N^2)$).
- File transfer time = $F / \text{throughput}$.

NUMBERS FROM SLIDES

- One-hop example: $L = 10$ Kbits, $R = 100$ Mbps, delay = 0.1 msec.
- Caravan 1: 62 minutes. Caravan 2: 7 min, 3 cars still at booth 1.
- 1 Gb/s link, 100 Mb/s per user, 10% active: circuit = 10 users, packet = 35 users with overload probability about 0.0004.
- Cable: 40 Mbps to 1.2 Gbps down, 30 to 100 Mbps up. DSL: 24 to 52 Mbps down, 3.5 to 16 Mbps up.
- WiFi: 11, 54, 450 Mbps, about 100 ft. Cellular: tens of km, tens of Mbps. Ethernet at home: 1 Gbps.
- Cat 5: 100 Mbps and 1 Gbps. Cat 6: 10 Gbps. Fiber: 10s to 100s Gbps.
- Geostationary satellite: about 270 msec delay. Starlink: under 100 Mbps down. Microwave: 45 Mbps.
- ARPAnet: 15 nodes (1972), 200 nodes (1979). About 15 billion devices in 2023.

LISTS TO MEMORIZE

- Three parts of the Internet: edge, access network, core.
- Two core functions: routing (global, planning) and forwarding (local, switching).
- Four delays: processing, queueing, transmission, propagation.
- Five layers: application, transport, network, link, physical.
- Data unit names: message, segment, datagram, frame.
- OSI extra layers: presentation and session.
- Attacks: sniffing, IP spoofing, DoS (botnet).
- Defenses: authentication, encryption (confidentiality), integrity checks (digital signatures), VPNs, firewalls.
- Cerf and Kahn principles: minimalism and autonomy, best-effort, stateless routing, decentralized control.
- Multiplexing in circuit switching: FDM (frequency bands) and TDM (time slots).
- Cable vs DSL: shared cable (HFC, headend, CMTS) vs dedicated telephone line (central office, DSLAM).
- Guided media: twisted pair, coax, fiber. Unguided: radio.

- Packet vs circuit switching: packet switching is shared, bursty-friendly and can lose packets. Circuit switching is reserved, guaranteed and wasteful when idle.

FIN !